



SQL Data Analysis Project



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Project Objective

- Analyze ecommerce business data using SQL
- Extract business insights
- Support decision making



Dataset Overview

- 1200 records
- Customer transactions
- Products
- Payments
- Revenue
- Referral Source



Business Questions Solved



- Revenue Analysis
- Product Analysis
- Customer Analysis
- Payment Analysis
- Marketing Analysis

Product Performance Analysis

```
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables
SELECT PRODUCT,
       round(SUM(TotalPrice), 2) AS Revenue
FROM df
GROUP BY Product
ORDER BY Revenue DESC
---
```

```
result2 = pysqldf(query2)

result2
```

[22] ✓ 0.0s

	Product	Revenue
0	Chair	195620.11
1	Printer	195612.61
2	Laptop	192126.56
3	Tablet	186568.95
4	Monitor	175651.41
5	Desk	167459.93
6	Phone	151722.39

```
notebook > sql_queries.py > M4 SQL Data Analysis Project > M4 Query 8: Products by Quantity Sold > M4 Business Question >
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline ...
SUM(Quantity) AS Total_Quantity_Sold
FROM df
GROUP BY Product
ORDER BY Total_Quantity_Sold DESC
---
```

```
result8 = pysqldf(query8)

result8
```

[22] ✓ 0.1s

	Product	Total_Quantity_Sold
0	Chair	562
1	Printer	542
2	Laptop	535
3	Desk	508
4	Tablet	497
5	Monitor	480
6	Phone	411

Payment behavior & Revenue Analysis

notebook > sql_queries.py > M4 SQL Data Analysis Project > M4 Query 4: Orders by Payment Method > M4 Business Question > query4

```
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[10] SELECT
    PaymentMethod,
    COUNT(*) AS Total_Orders
FROM df
GROUP BY PaymentMethod
ORDER BY Total_Orders DESC
'''

result4 = pysqldf(query4)

result4
```

	PaymentMethod	Total_Orders
0	Online	258
1	Cash	246
2	Credit Card	234
3	Debit Card	232
4	Gift Card	230

notebook > sql_queries.py > M4 SQL Data Analysis Project > M4 Query 7: Revenue by Payment Method > M4 Business Question > query7

```
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[10] query7 = '''
SELECT
    PaymentMethod,
    ROUND(SUM(TotalPrice),2) AS Revenue
FROM df
GROUP BY PaymentMethod
ORDER BY Revenue DESC
'''

result7 = pysqldf(query7)

result7
```

	PaymentMethod	Revenue
0	Credit Card	263847.63
1	Online	262442.94
2	Cash	259786.29
3	Gift Card	246323.92
4	Debit Card	232361.18

Customer Revenue Contribution

```
notebook > sql_queries.ipynb > M4 SQL Data Analysis Project > M4 Query 6: Top Customers by Revenue > M4 Business Question > query6 = """  
Generate + Code + Markdown | Run All Restart Clear All Output Jupyter Variables Outline  
ORDER BY Revenue DESC  
LIMIT 10  
""  
results6 = pysqldf(query6)  
results6  
[10] ✓ 0.0s  
---  
CustomerID Revenue  
0 C38840 5723.23  
1 C57276 3456.40  
2 C67260 3390.80  
3 C13877 3384.90  
4 C18404 3370.20  
5 C16775 3353.75  
6 C65986 3352.40  
7 C47778 3334.00  
8 C59183 3322.55  
9 C25276 3313.90
```

Order Lifecycle & Revenue Impact

```
notebook > sql_queries.py > M4 SQL Data Analysis Project > M4 Query 5: Order Status Analysis > M4 Business Question > query5 = ""
```

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```
> query5 = """
SELECT
    OrderStatus,
    COUNT(*) AS Total_Orders
FROM df
GROUP BY OrderStatus
ORDER BY Total_Orders DESC
"""

result5 = pysqldf(query5)

result5
```

[18] ✓ 0.0s

	OrderStatus	Total_Orders
0	Cancelled	250
1	Returned	247
2	Pending	237
3	Shipped	235
4	Delivered	231

```
notebook > sql_queries.py > M4 SQL Data Analysis Project > M4 Query 9: Revenue by Order Status > M4 Business Question > query9 = ""
```

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```
> query9 = """
SELECT
    OrderStatus,
    ROUND(SUM(TotalPrice),2) AS Revenue
FROM df
GROUP BY OrderStatus
ORDER BY Revenue DESC
"""

result9 = pysqldf(query9)

result9
```

[17] ✓ 0.0s

	OrderStatus	Revenue
0	Cancelled	276396.21
1	Pending	256328.15
2	Shipped	246159.58
3	Returned	243277.70
4	Delivered	242600.32

Marketing Channel Performance

```
notebook > sql_queries.ipynb > M SQL Data Analysis Project > M Query 10: Referral Source Performance > M Business Question > query10 = ""
```

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```
query10 =  
SELECT  
    ReferralSource,  
    COUNT(*) AS Total_Customers  
FROM df  
GROUP BY ReferralSource  
ORDER BY Total_Customers DESC  
""
```

result10 = pysqldf(query10)

result10

[23] ✓ 0.0s Python

	ReferralSource	Total_Customers
0	Instagram	259
1	Email	250
2	Google	241
3	Facebook	228
4	Referral	222

Conclusion

- Identified top-performing products
- Identified high-value customers
- Identified strongest marketing channels
- Generated actionable business recommendations

